AYMANE AIT DADS

Artificial Intelligence / Machine Learning Intern | Computer Vision · Foundation Models · Data Science

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT on structured multi-category reasoning tasks. Built and deployed a full data science pipeline at Orange Maroc on 100K+ fiber-optic network records, reducing manual analysis time by ~60% and mapping customer performance profiles to 5 commercial service tiers using Random Forest and K-Means. Brings hands-on experience in computer vision, NLP, foundation model fine-tuning, and scalable data analytics directly aligned with Bosch RTC-NA research in Foundation Models, Visual Analytics, and Industry 4.0 AI solutions. Aims to contribute to trustworthy, scalable AIoT research at Bosch by applying rigorous ablation methodology and multi-domain ML engineering across industrial sensor, image, and language data.

CORE COMPETENCIES

Foundation Models | Computer Vision | Natural Language Processing | Data Science & Analytics | Explainable AI (XAI) | Scalable AI Solutions | Industry 4.0 / IoT | Big Data Visual Analytics

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a **Random Forest + K-Means pipeline** on 100K+ fiber-optic network records (upload speed, latency, jitter) to classify network quality across thousands of nodes for the network optimization team.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with outputs directly consumed by network engineers to drive infrastructure maintenance decisions.
- Reduced manual data analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible Python scripts built on Pandas and Scikit-learn.
- Delivered **Power BI stakeholder presentations** translating model outputs and cluster diagnostics into actionable maintenance recommendations for non-technical business audiences.

PROJECTS

AI-Generated Image Detection - NTIRE 2026 @ CVPR *EURECOM ImSecu + International Benchmark · 2025 · Ranked 1st in Class*

- Ranked **1st in EURECOM class leaderboard** (0.791 AUC) and submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head over 250K images from 25+ generator types.
- Achieved optimal out-of-distribution generalization by applying **dual learning rates** (2e-6 CLIP backbone, 5e-4 head), 10-view TTA ensemble, and FP16 mixed precision - finding 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC).

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

NVIDIA Nemotron Model Reasoning Challenge *Kaggle Competition · In Progress (June 2026) · Top 17 / 3,000+ Teams*

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · PEFT · TRL · vLLM · Hugging Face Transformers · Triton

Anomalous Sound Detection - Industrial Equipment *EURECOM · 2025 · Predictive Maintenance*

- Designed an unsupervised fault detection system for slide-rail industrial machinery using a **Transformer autoencoder on Mel spectrograms** with SpecAugment, achieving AUC 0.80 with no labeled anomaly data.
- Demonstrated applicability to **Industry 4.0 IoT sensor analytics** by building a fully unsupervised pipeline reproducing real-world conditions where labeled fault examples are unavailable at deployment.

PyTorch · Mel Spectrograms · SpecAugment · Transformer Autoencoder

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Statistics, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya *2021 - 2023*

SKILLS

Data Science & Analytics: Python | Pandas | NumPy | Scikit-learn | Power BI | SQL | Feature Engineering

AI / ML & Foundation Models: PyTorch | Hugging Face Transformers | LoRA / PEFT | OpenCLIP | ViT | vLLM | TRL

MLOps & Infrastructure: Git | Linux | FP16/BF16 Mixed Precision | CUDA | Docker | Ablation Studies | Kaggle GPU